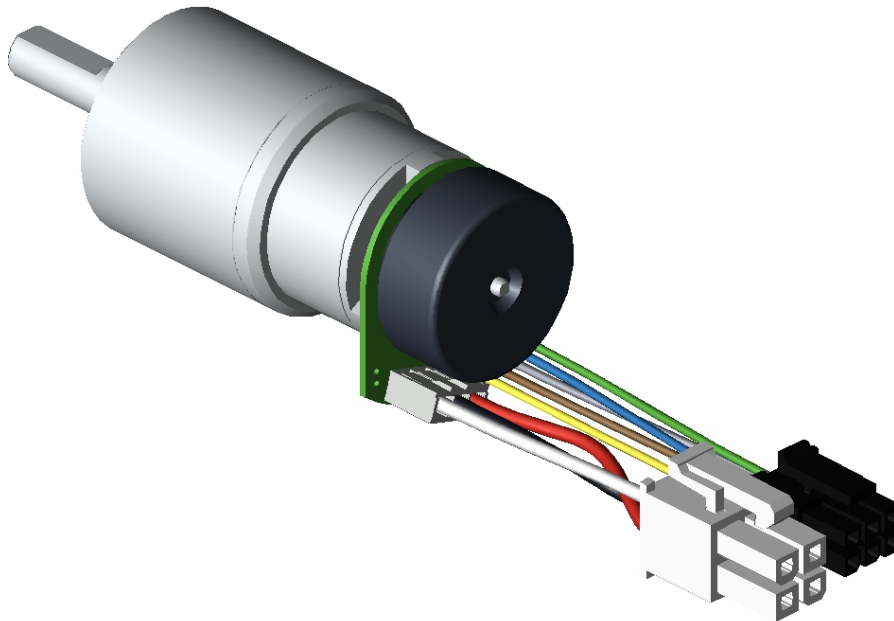


Configured drive

Motor - ECXFL22S KL A STD 18V
Gearhead - GPX26 A 138:1

Part number: B82ED3D50576 Revision number 1

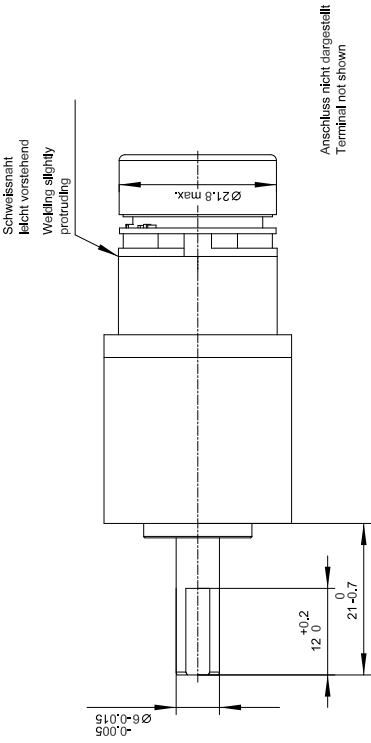
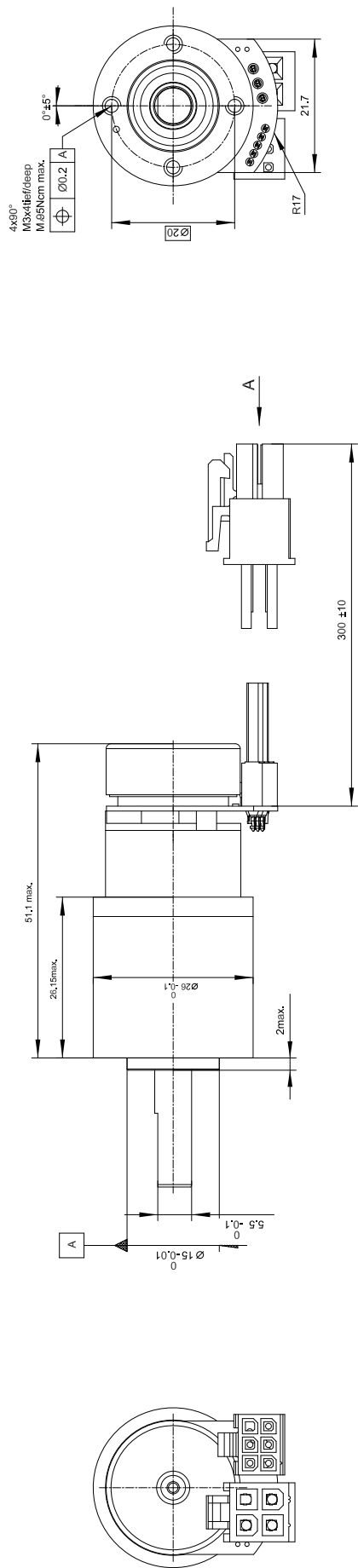
General Terms and Conditions https://www.maxongroup.com/maxon/view/content/terms_and_conditions_page



To open the integrated CAD file, please save this document and open it in Acrobat Reader. The STEP file is available after a double-click on the pin icon.

B82ED3D50576.stp (STP AP 214)

Open configuration: <https://www.maxongroup.com/maxon/view/configurator/&ConfigID=B82ED3D50576>



Unit of measure: mm



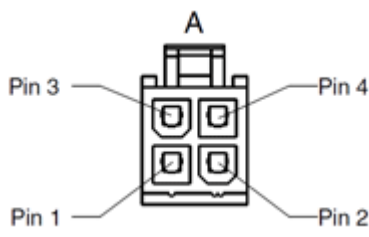
ISO 5456-1

ISO 1101

ISO 965-1

ISO 2768-m

ISO 8015

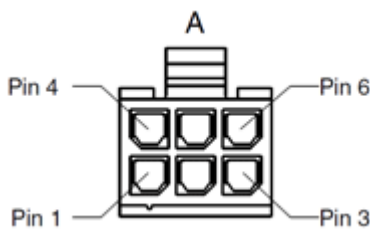


Connector type, motor

Molex 39-01-2040

Pin 1	Motor 1	Red
Pin 2	Motor 2	Black
Pin 3	Motor 3	White
Pin 4	NC	

Cable type: AWG22



Connector type, motor

Molex 43025-0600

Pin 1	Hall-Sensor 1	Yellow
Pin 2	Hall-Sensor 2	Brown
Pin 3	Hall-Sensor 3	Gray
Pin 4	GND	Blue
Pin 5	+2.5...5.5 V DC	Green
Pin 6	NC	

Cable type: AWG28

Summary of your selected configuration

Total weight of the drive: 138.8 g

ECXFL22S KL A STD 18V

Product detail		
Commutation		A
Nominal voltage		18
Motor bearings		Preloaded ball bearing
Version		STD
Elec. connection, motor		
Electrical connection, motor		Cable with connector
Connector type, motor		Molex 39-01-2040

GPX26 A 138:1

Product detail		
Gearhead type		AA
Reduction		138
Number of stages		3

Legend for part designation

EC-Motor:		DC-Motor:		IDX Drive/Motor:	
A	Hall sensor version	CLL	Capacitor long life (spark suppression)	AB	Integrated brake
B	Sensorless version	DC-max	DC motors (brushed)	CO	CANopen interface
BL	Brushless	DCX	DC motors (brushed)	ENC	Encoder
C	Integrated electronics	EB	Precious metal brushes	ET	EtherCAT interface
ECX	EC-motor (brushless)	GB	Graphite brushes	IDX	Drive System
FL	FLAT (type series name)	KL	Ball bearings	IO	I/O interface
HP	High Power	S/M/L	Short / Medium / Long	P	Positioning-control
HTQ	High Torque	SL	Sintered sleeve bearings	S	Speed-control
HTQF	High Torque, fan-ventilated			S/M/L	Short / Medium / Long
KL	Ball bearings				
M/L/X	Medium / Long / Extra long				
PR	PRIME (type series name)				
SP	SPEED (type series name)				
STD	Standard				
STE	Sterilizable				
STEC	Sterilizable, ceramic bearing				
TQ	TORQUE (type series name)				
Gearhead		Encoder/Sensor:			
A	Standard version	ABS	Absolute		
C	Ceramic version	BiSSC	Bidirectional serial synchronous (protocol)		
GPX	Planetary gearhead	COMM	Commutation signals		
HP	High Power	EASY	(type series name)		
LN	Reduced noise level	EMT	(type series name)		
LZ	Reduced backlash	ENX	Encoder		
SP	SPEED (type series name)	FFC	Flexible flat cable		
ST	Number of stages	GAMA	(type series name)		
STE	Sterilizable	IMP	Pulses		
STES	Sterilizable, with sealing	INT	integrated		
UP	Ultra Performance	MAG	(type series name)		
		QUAD	(type series name)		
		RIO	(type series name)		
		SSI	Synchronous serial Interface (protocol)		
		XT	Extended version		

ECXFL22S KL A STD 18V

Product specification

Values at nominal voltage

Nominal voltage	18 V
No load speed	10600 min ⁻¹
No load current	32.7 mA
Nominal speed	5640 min ⁻¹
Nominal torque (max. continuous torque)	7.92 mNm
Nominal current (max. continuous current)	0.511 A
Stall torque calculated without saturation effect	23.3 mNm
Stall current	1.55 A
Max. efficiency	67.453 %

Characteristics

Max. output power continuous	W
Terminal resistance phase-to-phase	11.6 Ω
Terminal inductance phase-to-phase	2.18 mH
Torque constant	15.7 mNm A ⁻¹
Speed constant	610 min ⁻¹ V ⁻¹
Speed/torque gradient	453 min ⁻¹ mNm ⁻¹
Mechanical time constant	31.8 ms
Rotor inertia	6.7 gcm ²

Thermal data

Thermal resistance housing-ambient	7.51 KW ⁻¹
Thermal resistance winding-housing	9.57 KW ⁻¹
Thermal time constant of the winding	6.49 s
Thermal time constant of the motor	18 s
Ambient temperature	-40...100 °C
Max. winding temperature	125 °C

Mechanical data

Max. permissible speed	15000 min ⁻¹
Axial play	0...0.14 mm
Preload	1.5 N
Direction of force	
Radial backlash	0 mm
Max. axial load (dynamic)	1.8 N
Max. force for press fits (static)	26 N
Static, supported shaft	200 N
Max. radial load 5 mm from flange	3.5 N
Measurement from the flange	5 mm

Further specifications

Number of pole pairs	6
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Number of phases	3
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More information on the motor: <https://www.maxongroup.com/maxon/view/product/ECXA22SZF10R6IL1Y658A?download=show>

GPX26 A 138:1

Product specification

Gearhead data

Reduction	138:1
Absolute reduction	341550/2470
Number of stages	3
Max. continuous torque	4.50 Nm
Max. intermittent torque	6.2 Nm
Direction of rotation, drive to output	=
Max. efficiency	75 %
Average backlash no-load	1.1 °
Mass inertia	1.282 gcm ²
Max. transmittable power (continuous)	12 W
Max. short-time transferable output	15 W

Technical data

Output shaft bearing	
Max. radial play, 5 mm from flange	max. 0.1 mm
Axial play	0...0.4 mm
Max. permissible radial load, 10 mm mm from flange	150 N
Max. permissible axial load	80 N
Max. permissible force for press fits	120 N
Max. continuous input speed	10000 min ⁻¹
Max. intermittent input speed	12500 min ⁻¹
Recommended temperature range	-40..100 °C

More information on the gearhead: <https://www.maxongroup.com/maxon/view/product/GPX26AAKLSL0138CPLW?download=show>